

An Ultrasound Imaging System With On-Chip Per-Voxel RX Beamfocusing for Real-Time Drone Applications

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Abstract—For drone vision and navigation, low-power 3-D depth sensing with robust operations against strong/weak light and various weather conditions is crucial. CMOS image sensor (CIS) and light detection and ranging (LiDAR) can provide high-fidelity imaging. However, CIS lacks depth sensing and has difficulty in low light conditions. LiDAR is expensive with issues of dealing with strong direct interference sources. Ultrasound imaging system (UIS), on the other hand, is robust in various weather and light conditions and is cost-effective. However, in air channel, it often suffers from long image reconstruction latency and low framerate. To address these issues, we present a UIS application-specific integrated circuit (ASIC) that adopts the one-shot transmitter (TX) and on-chip per-voxel receiver (RX) beamfocusing (PV-RXBF) image reconstruction scheme. The ASIC adopts the designs of fully differential charge-reuse high-voltage TX (FDCR-HVTX), digital back-end (DBE), and an on-chip power management unit (PMU). FDCR-HVTX generates 28 V_{pp} pulses and reduces the average power consumption by 25% by charge reuse (CR). The DBE achieves 7.76-μs processing latency and 9.83M-FocalPoint/s throughput to effectively translate real-time 3-D image streaming at 24 frames/s. A prototype UIS, with an 8 × 8 bulk piezo transducer array, is assembled with the proposed ASIC and a wireless data transmission module [field-programmable gate array (FPGA) + ESP32] on an entry-level consumer drone, and the real-time wireless 3-D image streaming at 24 frames/s

with a range of 7 m is verified while the drone is flying. The ASIC implemented in 180-nm 1P6M Standard CMOS occupies 32.5 mm² and consumes 142.3 mW.

Index Terms—3-D imaging, all light condition, charge reuse (CR), depth sensing, drone, high-voltage transmitter (TX), low power, per-voxel RX beamfocusing (PV-RXBF), real time, standard CMOS, ultrasound.

I. INTRODUCTION

UNMANNED aerial vehicles (UAVs), or drones, are gaining popularity in recent years. Some consumer-grade drones can fly up to 20 m/s, and 3-D depth-sensing systems for such UAVs should have low latency, sufficient framerate, and low power consumption. As a safety measure, the low latency and high framerate are essential, especially for consumer models, as the operators are often less professionally trained. Unfortunately, due to the power, size, and cost considerations, many existing drones do not have a 3-D depth-sensing capability and/or are limited to only distance detection at the front side [1]. To address these issues, CMOS image sensors (CISs), light detection and ranging (LiDAR), and ultrasound image systems (UISs) could be utilized. Fig. 1 shows the comparison of these technologies. CISs [2], [3], [4], [5] offer the highest framerate with good image quality, but they are vulnerable to bright exposure and low light conditions. More importantly, it lacks direct depth information, making it suboptimal as the onboard safety sensor for drones. LiDAR